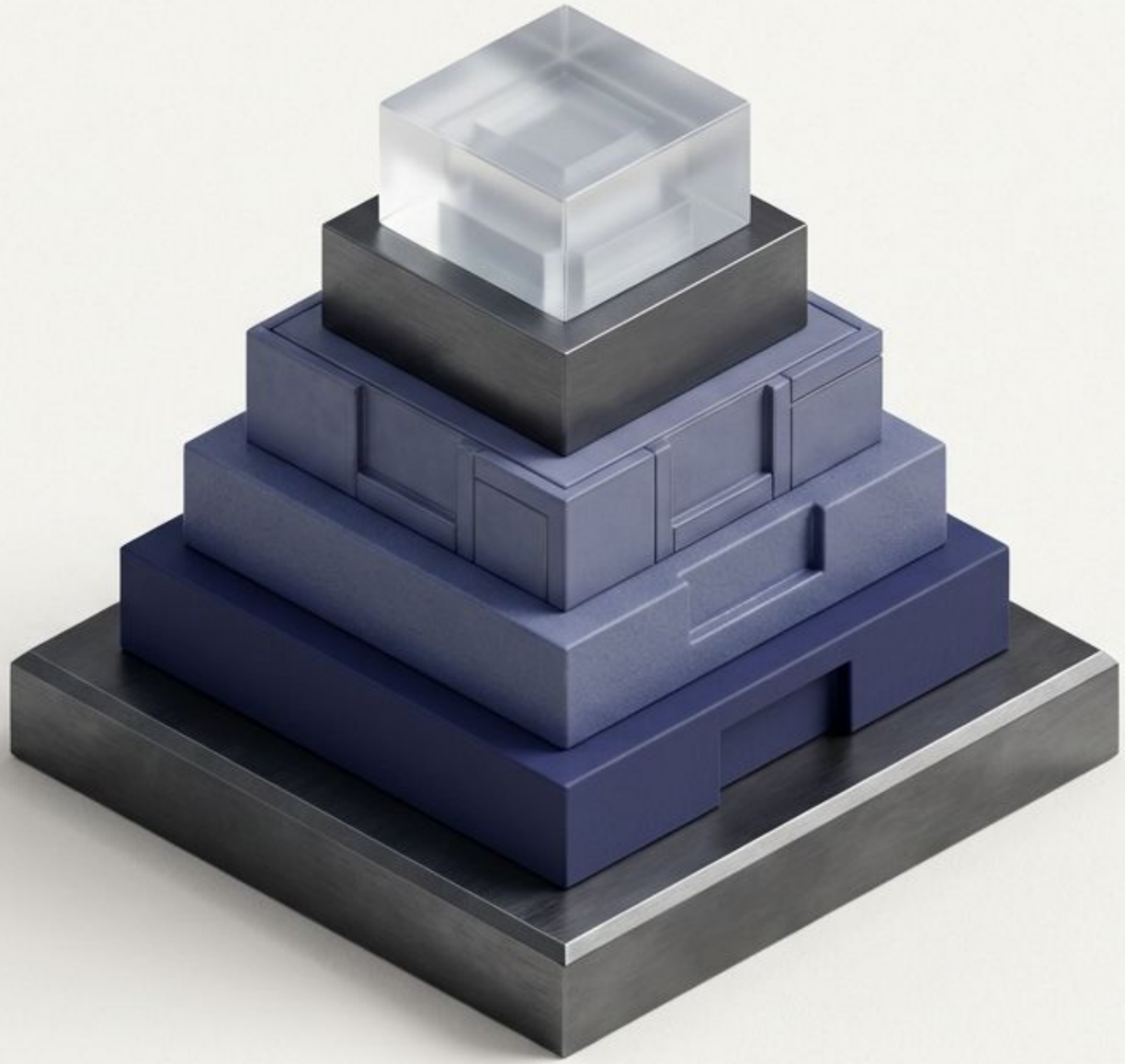
Abstractions for Multimedia Programming

Mastering the tech stack from bare metal to user interface

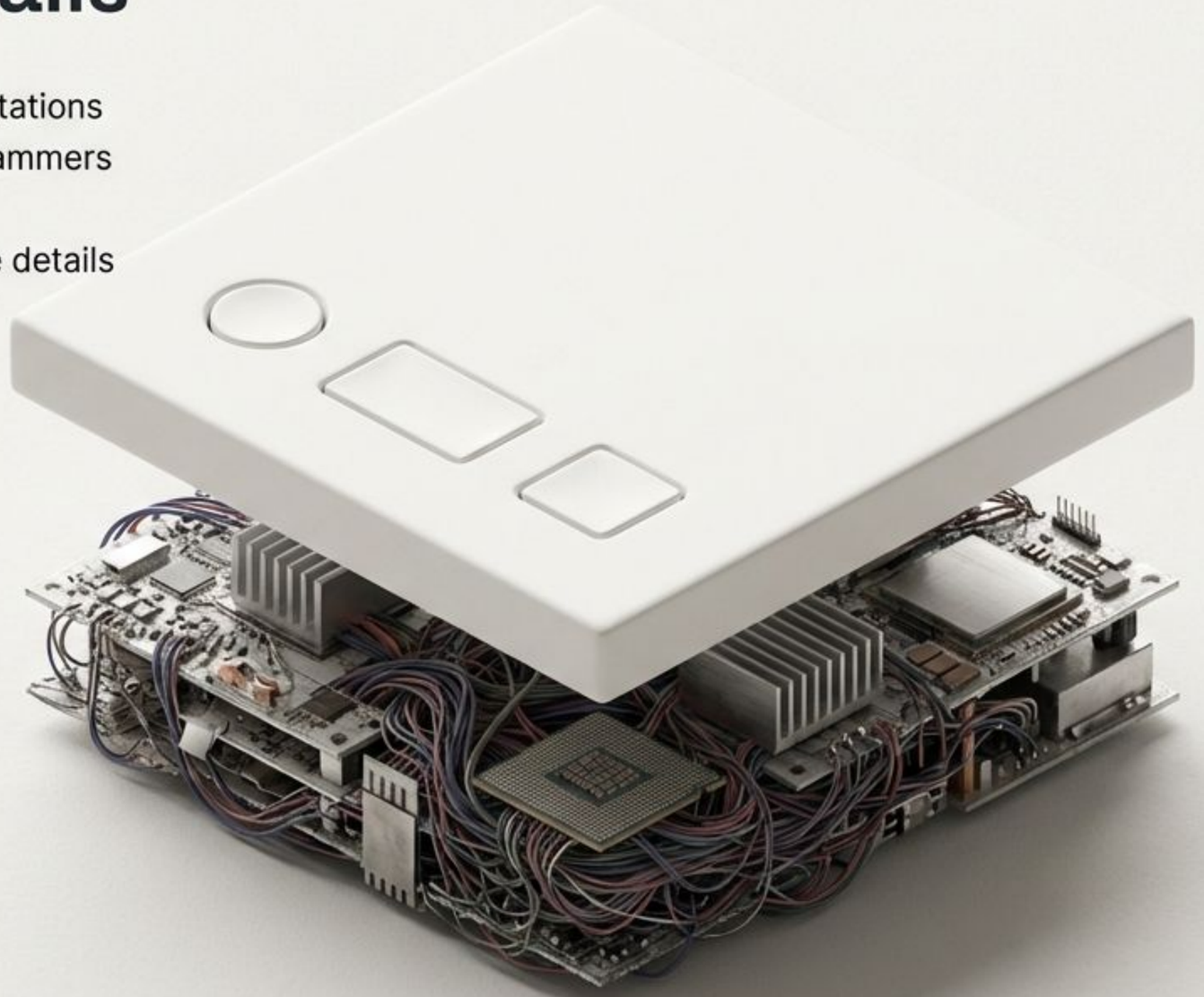
The architecture of modern multimedia systems



Prepared By : Er, Sharat Maharjan

Hiding the Messy Details

- Process of obscuring complex underlying implementations
- Provides a simplified, consistent interface for programmers
- Allows development at a higher conceptual level
- Removes the need to understand intricate hardware details



The 6 Levels of Multimedia Abstraction

6. Application Level
User-facing software

5. Toolkit / Framework Level
Integrated tools

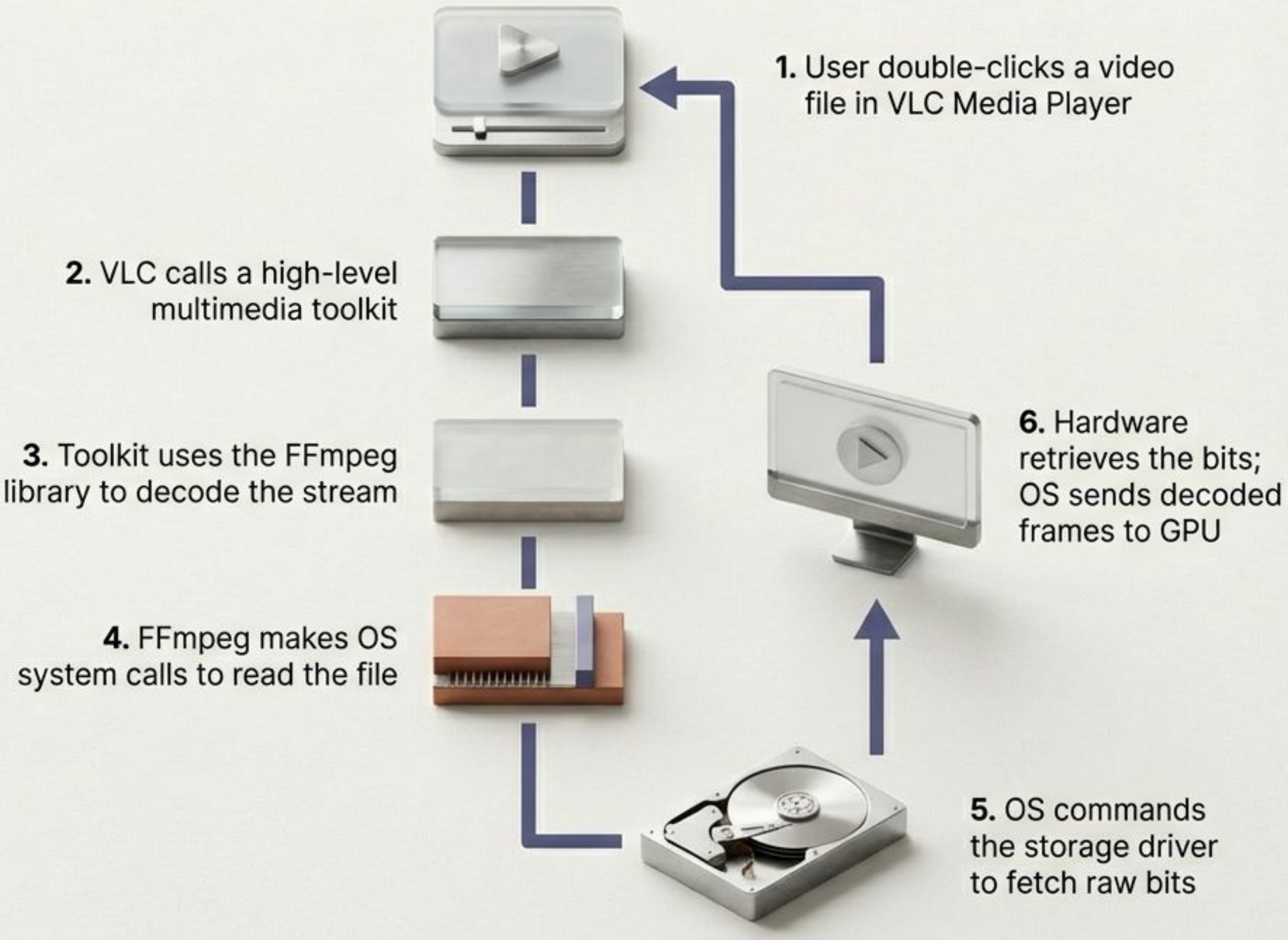
4. Library Level
Pre-written code collections

3. Operating System Level
System calls & APIs

2. Firmware / Driver Level
Direct hardware control

1. Hardware Level
Physical CPUs, GPUs, sensors

The Anatomy of Playing an MP4



The Operating System Foundation

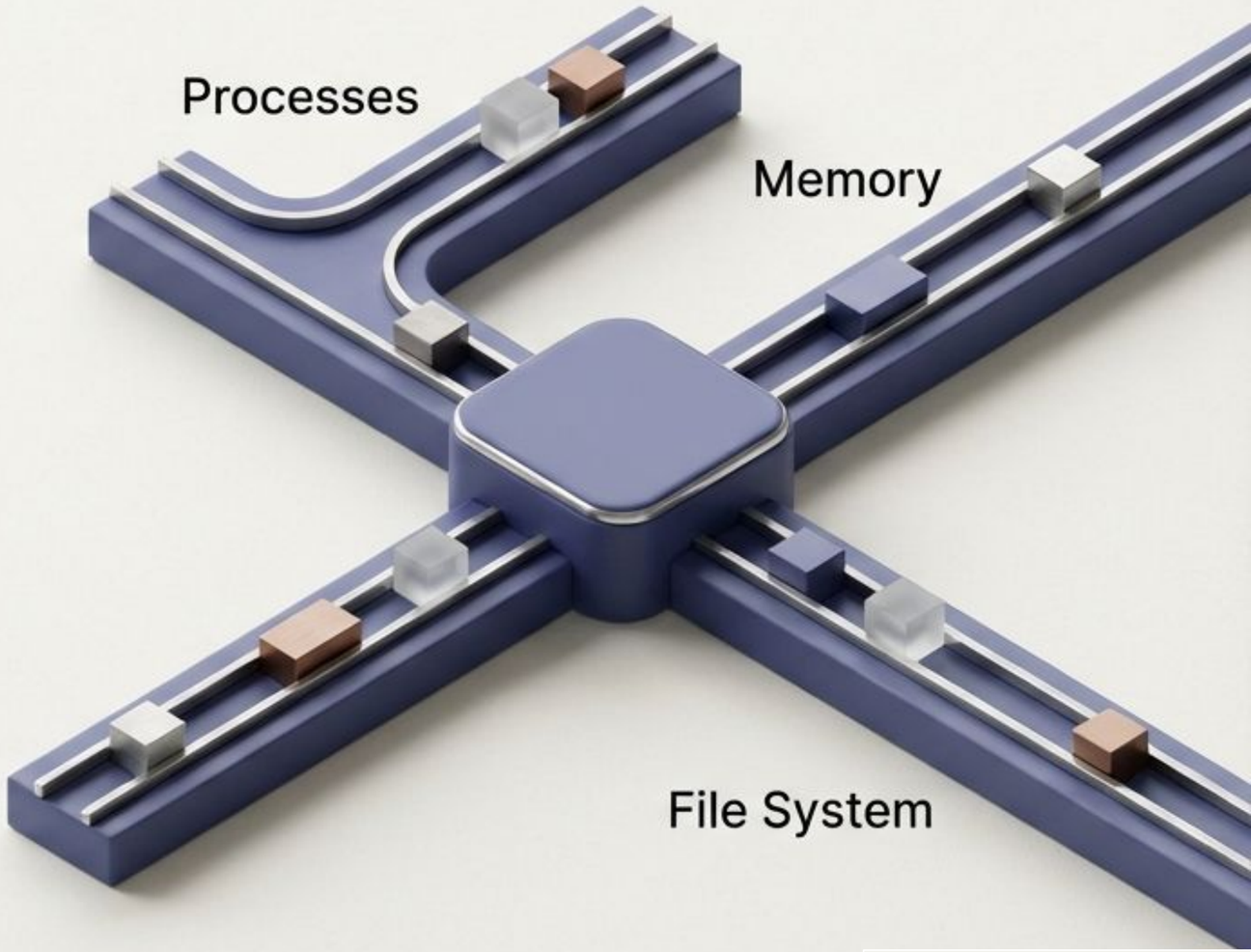
Core system software managing hardware resources

Process Management: Runs multiple apps simultaneously

Memory Management: Allocates RAM for massive video files

File System Management: Organizes and retrieves media

Provides built-in multimedia APIs (e.g., Core Audio)



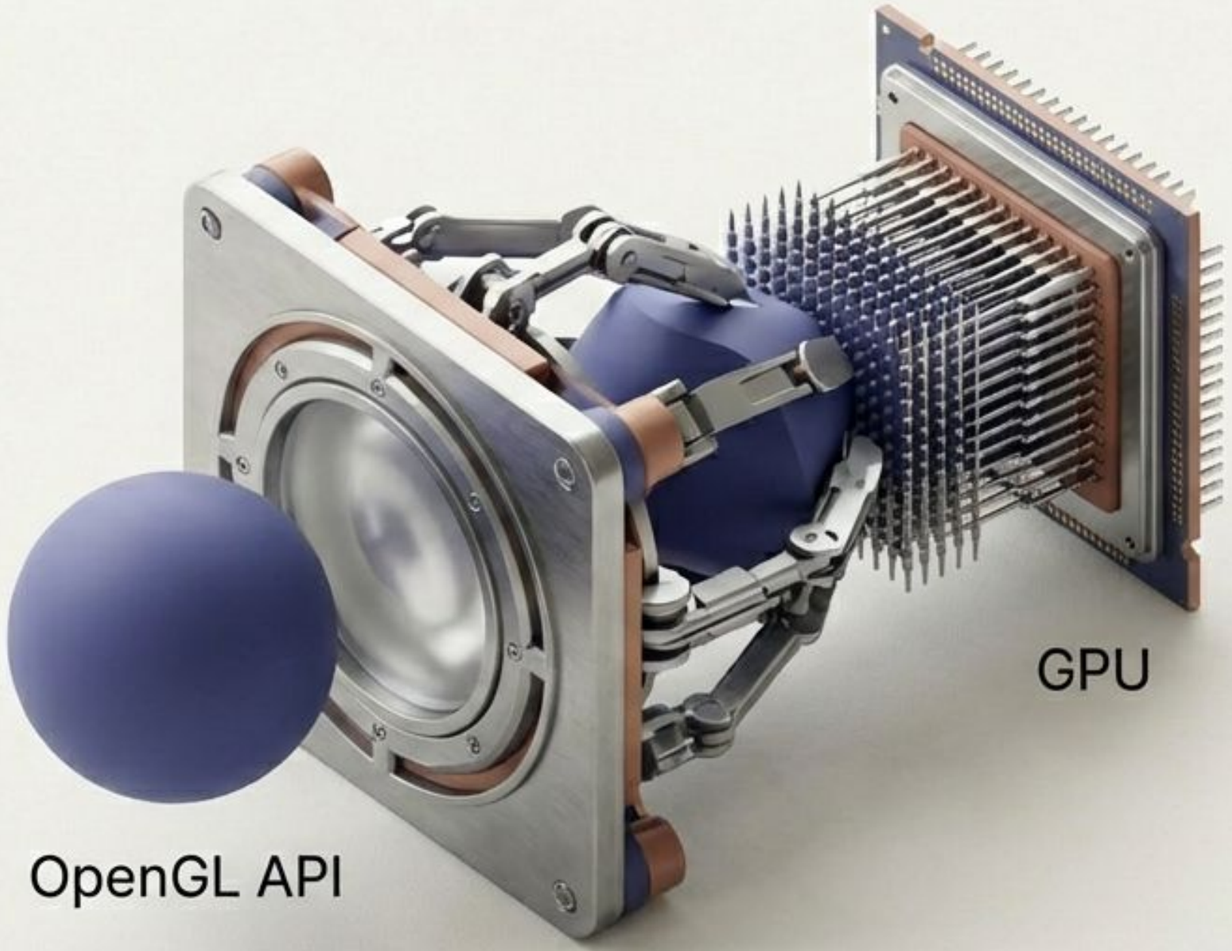
Translating Software to Hardware

Device Drivers bridge the OS and physical components

Translate standardized API commands into hardware instructions

Crucial for graphics cards, sound cards, and webcams

Example: Translating OpenGL commands for specific GPUs



Libraries: Reusable Code Blocks

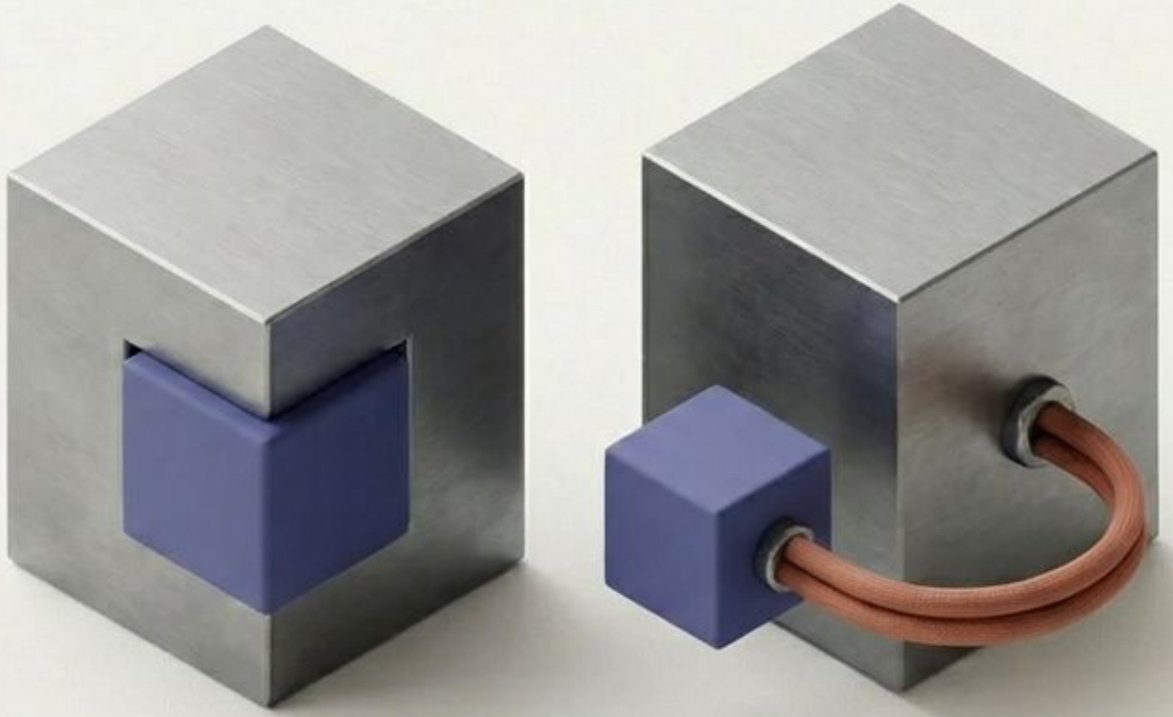
Collections of pre-compiled, optimized code

Shield developers from complex, low-level algorithms

Static Libraries: Linked at that executable. Linked at compile-time directly into the executable

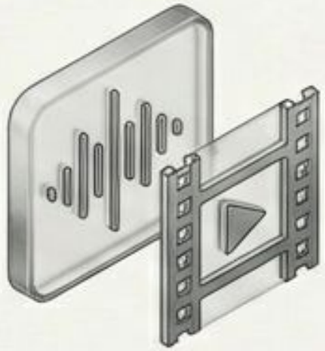
Dynamic/Shared Libraries: Linked at run-time (e.g., .dll, .so)

Can be shared simultaneously by multiple programs

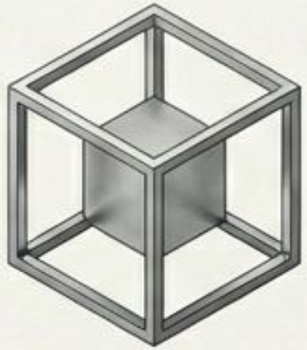


Industry Standard Libraries

Images:
libjpeg, libpng



Audio/Video:
FFmpeg



Graphics: OpenGL, DirectX, Vulkan

Advantages

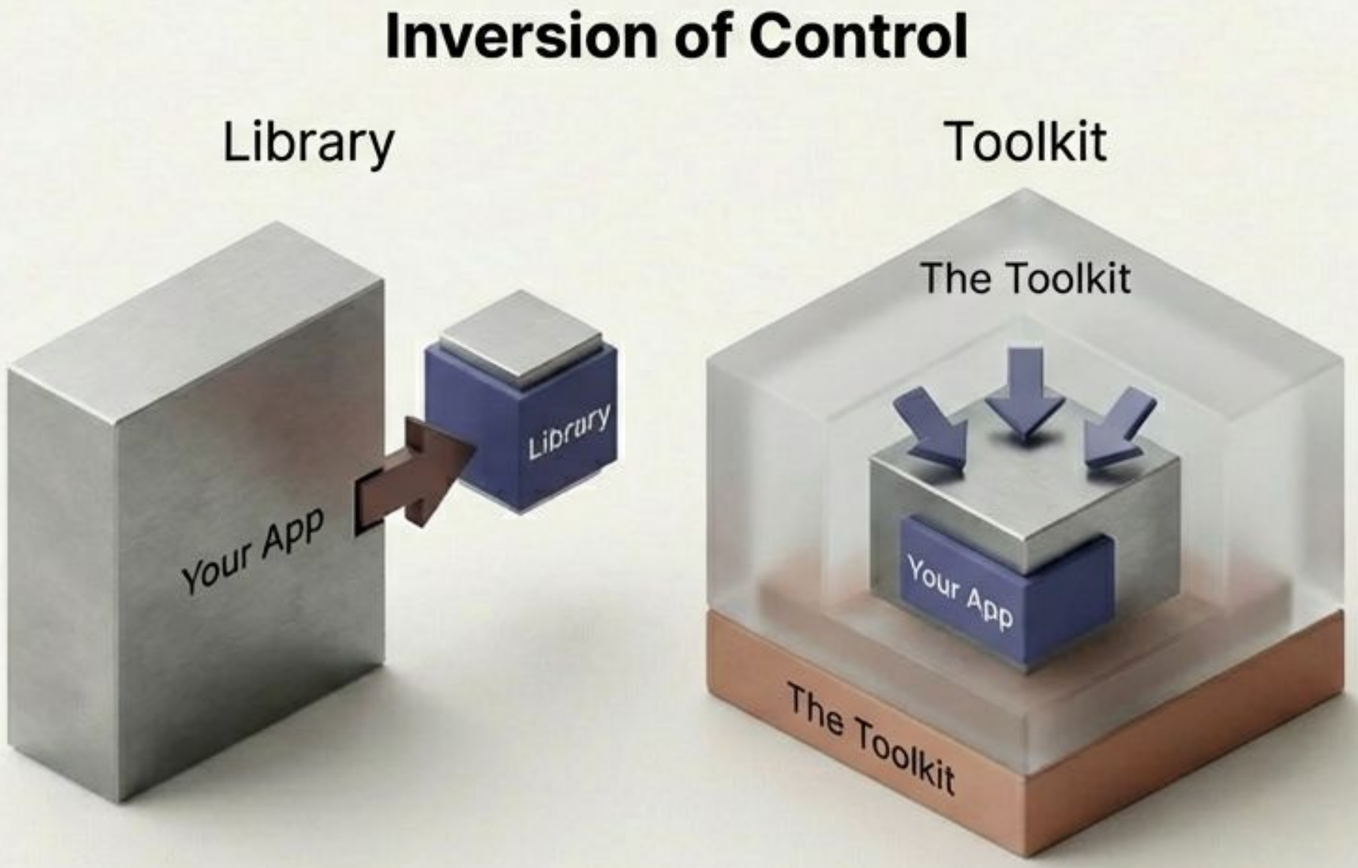
High reusability, reliability, and portability

Disadvantages

Strict learning curves, versioning conflicts (DLL Hell)

Comprehensive Toolkits and Frameworks

- Highly integrated environments defining application structure
- Inversion of Control: Don't call us, we'll call you
- Dictates the flow of control rather than just providing functions
- Abstracts away operating system differences entirely



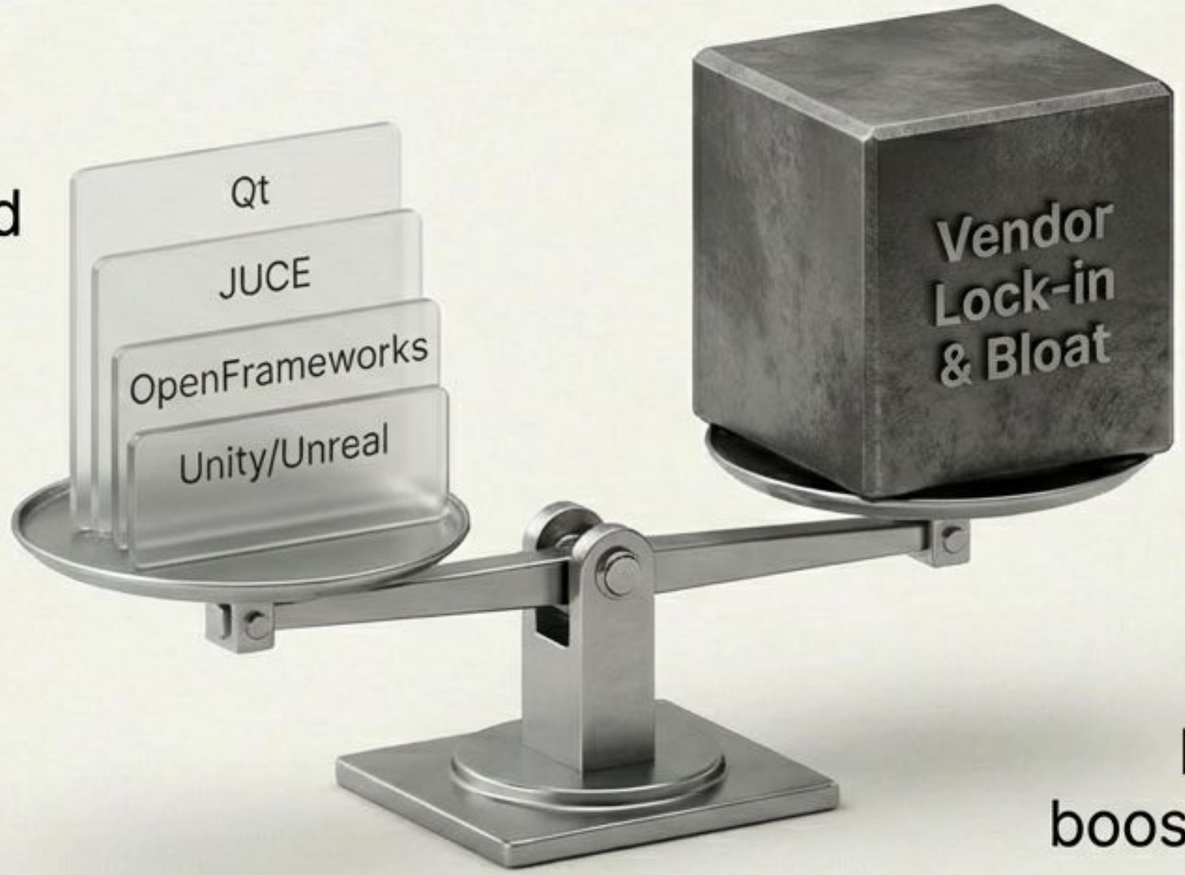
Heavyweight Framework Ecosystems

Qt: Cross-platform GUI and multimedia

JUCE: Industry standard for audio apps and VST plugins

OpenFrameworks: C++ toolkit for creative coding and installations

Unity / Unreal Engine: Massive toolkits for interactive 3D/2D



Trade-offs:

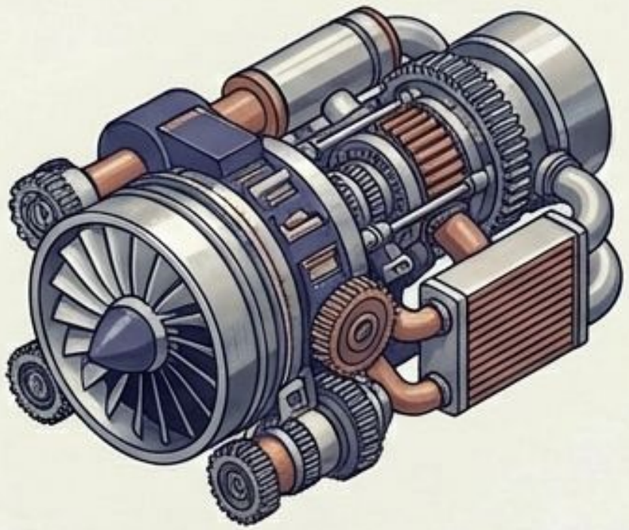
Dramatic speed boosts vs. Bloat and Vendor Lock-in

High-Level Languages: Power vs Speed

Provide strong abstractions away from machine architecture

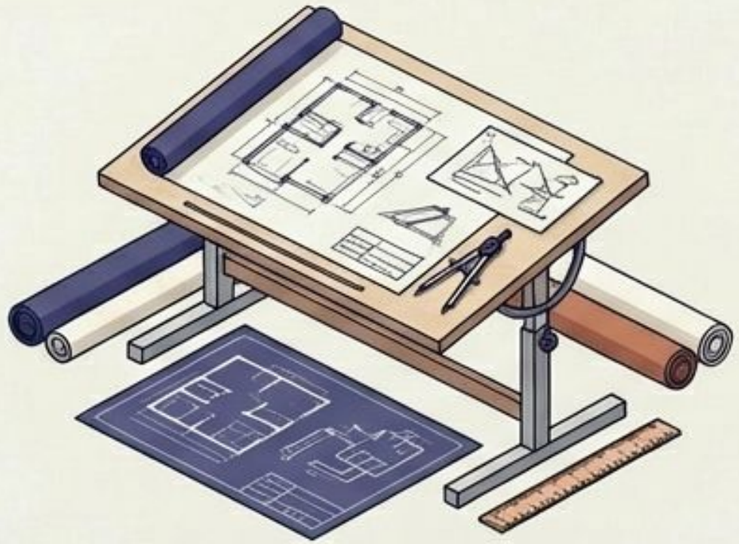
C/C++

- Dominant for performance-critical engines and core libraries
- Provides fine-grained memory and hardware control



Python

- Champion of rapid prototyping and scripting
- Leverages extensive libraries despite slower execution



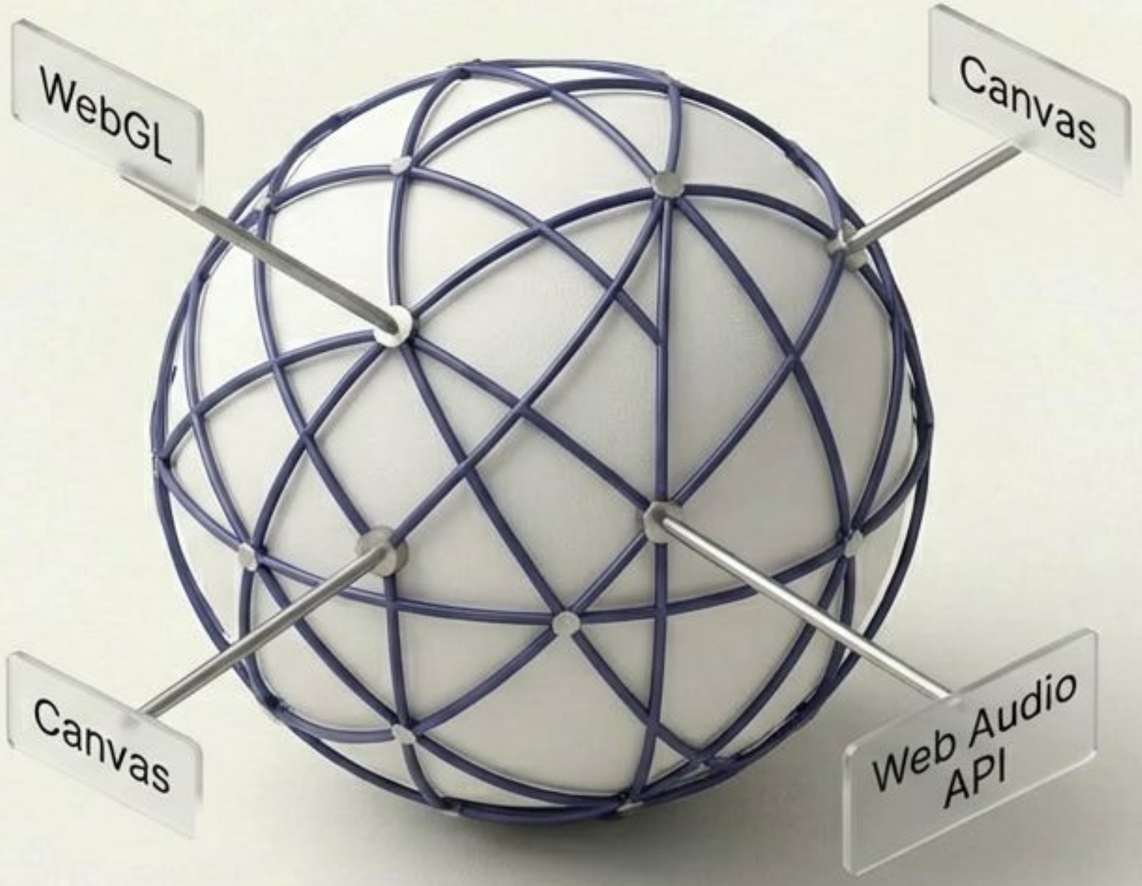
Web and Cross-Platform Execution

JavaScript: The undisputed language of the web

- Drives plugin-free multimedia via Web Audio API, WebGL, and Canvas
- Essential for interactive visualizations and streaming services

Java: Utilized for cross-platform desktop and legacy Android systems

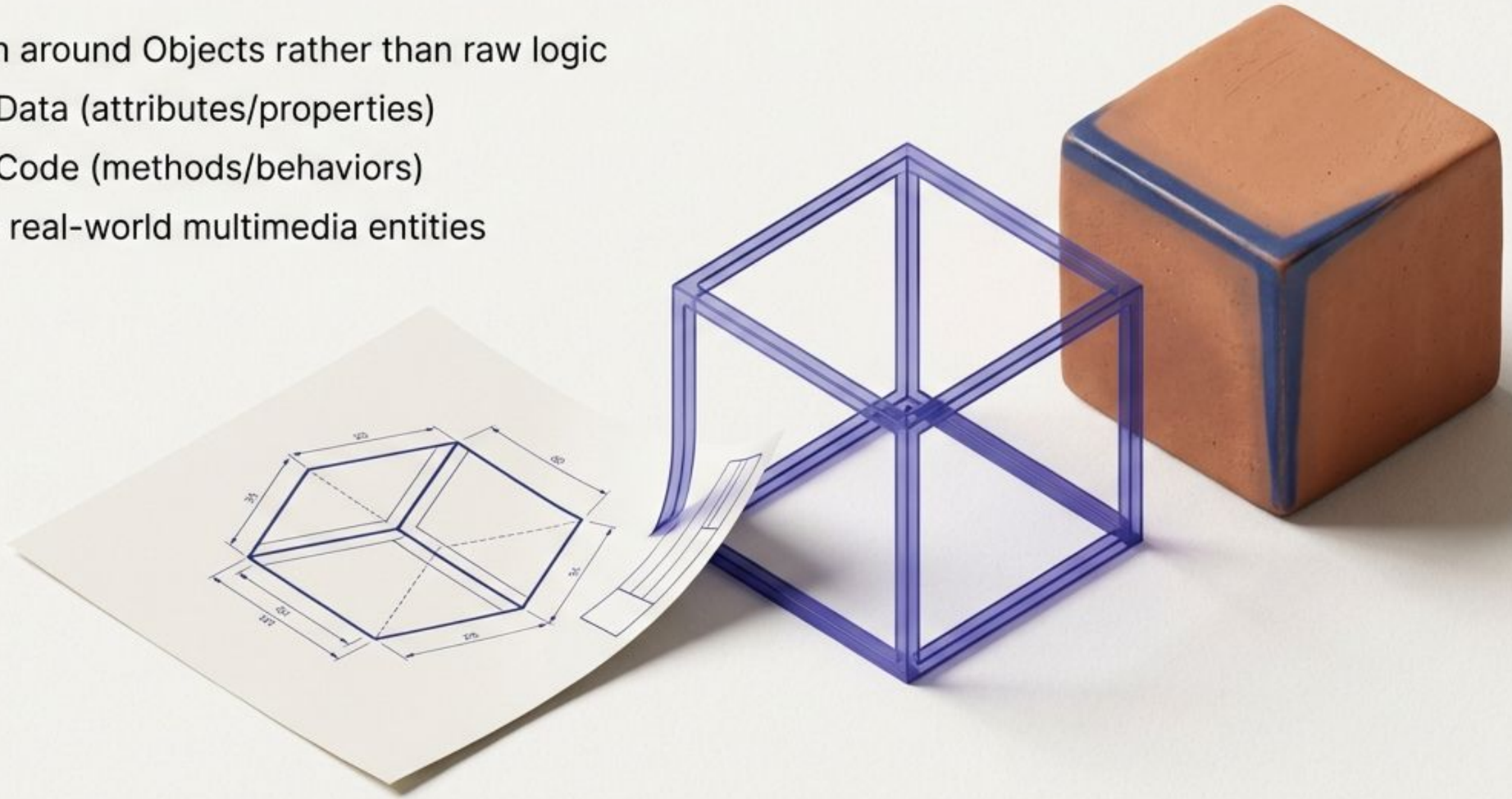
- Relies on bindings for lower-level libraries like OpenGL



Modeling Media as Objects

Object-Oriented Programming (OOP): The ideal multimedia paradigm

- Organizes design around Objects rather than raw logic
- Objects contain Data (attributes/properties)
- Objects contain Code (methods/behaviors)
- Perfectly mirrors real-world multimedia entities



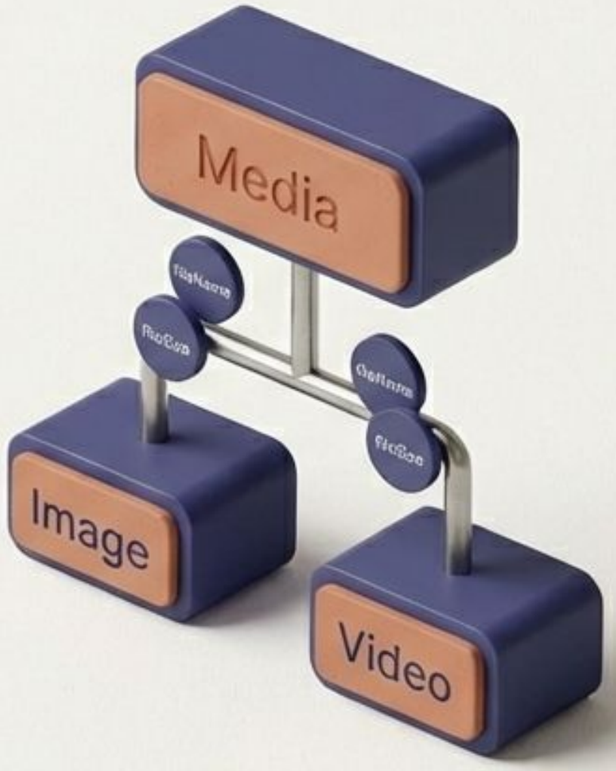
Encapsulation and Inheritance

Encapsulation: Bundling data and hiding internal mechanics

- Example: Calling AudioPlayer.play() hides complex MP3 decoding

Inheritance: Deriving new classes from existing base blueprints

- Example: Base class Media passes attributes to derived classes
- Derived classes Image and Video inherit fileName and fileSize



Polymorphism and The Tech Stack

Polymorphism: Different objects respond uniquely to identical commands

- Example: displayInfo() shows a thumbnail for Images, but a frame for Video
- Simplifies architecture across vast multimedia systems
- The Result: Complex hardware tamed by layered software abstractions